

Docking Universal - Reusable Exploratory Protocol Report

Report generator version 0.7.0

Target: 2R8N

Interpretive status: exploratory site selection without a bound-ligand pose-recovery control

1. Configured docking protocol

This section records the settings used for this exploratory study. No target-specific bound-ligand pose-recovery control was available.

Parameter	Configured value
Validation status	Site-guided exploratory protocol; not evaluated by bound-ligand control
Engine	QuickVina-W
Engine version	recorded when screening runs
Exhaustiveness	16
Modes per job	15
Energy range	8.0 kcal/mol
Independent seeds	5
Receptor	2R8N.pdbqt
Docking box	2R8N_pocket1.conf

Reusable protocol: 2R8N_site-guided-exploratory_qvinaw_2026-09-11.duprotocol

2. Exploratory pocket configuration

No bound-ligand pose-recovery control was available. In its place, this section records how fpocket-generated cavity hypotheses were filtered and which docking box was selected. The fpocket cavities shown in Figure 1 are reference evidence for review, not automatic proof of a binding site. This documents site selection, but it does not validate the biological site or the accuracy of docked poses.

Pocket configuration	Recorded value
Interpretive status	Exploratory; not evaluated by a bound-ligand pose-recovery control
fpocket eligibility threshold used	0.0 (relaxed sensitivity pass)
Top fpocket score	0.0724
Top-pocket druggability descriptor	0.0
Selected docking region	Box P1
Docking-box center (A)	49.029000, 35.415000, -14.441000
Docking-box dimensions (A)	26.0 x 26.0 x 26.0

Related-structure ligand evidence

Deposited ligands from the same protein, exact-sequence structures, and explicitly approved close structural homologs are treated as supporting ligand-location evidence. In the source lists, exact match means the same protein identifier or 100% sequence identity; partial match means an approved close structural homolog meeting the recorded thresholds. These sources do not independently validate a binding site or protocol, and orthology is not inferred. Ligands were aligned by their contacting protein chain and grouped by spatial overlap. Direct fpocket correspondence requires unique cavity contact and full ligand containment. Boxes start at 26 Å and expand within the recorded 36 Å dimension and 64,000 Å³ volume limits. P# denotes an fpocket-derived box; L# denotes ligand evidence without a matching cavity. Accepted alignments require identity 0.9, coverage 0.7, and C-alpha RMSD at most 2.0 Å. Ambiguous chain assignments were excluded. Excluded ambiguous chain assignments: 1ZTZ/CB5/P/1001.

Region	Evidence-based box recommendation
P2	Box: 26.0 × 26.0 × 26.0 Å; Direct support (fully contained by P2): Partial matches (close structural homolog): 1ZTZ/CB5, 3I8W/CB5. Additional ligand evidence at the same aligned site (supports the site location; not used to set Box P2 dimensions): Exact matches (100% sequence identity): 2R5P/MK1, 2R5Q/1UN; Partial matches (close structural homolog): 1D4Y/TPV, 1HPO/UNI, 1MRW/K57, 1MSM/JE2, 2PK5/075, 2PK6/O33, 3KDB/006, 3KDC/JZP, 3KDD/JZQ, 3NLS/O16, 3QBF/JHG.

Evidence review. Related-structure ligand locations were compared with fpocket candidates. The standard fpocket threshold did not provide sufficient correspondence, so candidates from a relaxed-sensitivity pass were retained. Selected for the protocol: Box P1. Selected automatically by fpocket rank. The selected box has no direct correspondence to the retained related-structure ligand evidence. No close structural-homolog evidence was used.

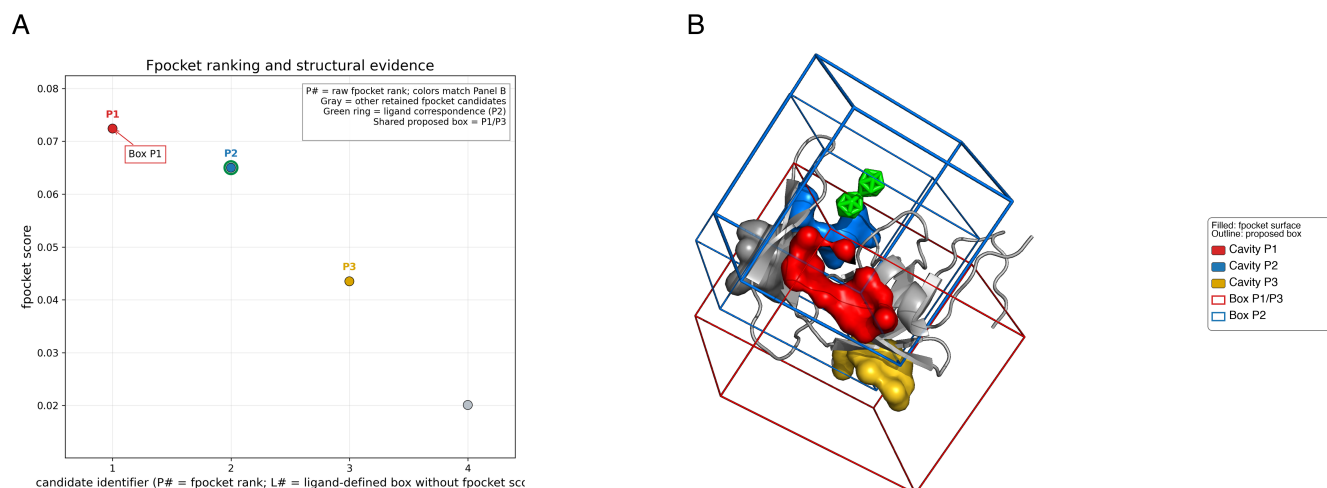


Figure 1. Exploratory pocket analysis. (A) Up to ten highest-ranked fpocket candidates; the top three are color matched to Panel B. Green rings mark direct correspondence with deposited ligand evidence; L# marks a ligand-defined box without a corresponding fpocket cavity. (B) Surfaces represent cavities; wireframes represent boxes. The same color-matched pocket surfaces and selected docking regions are shown on the receptor. The selected region is identified by its color-matched surface and box. Scores and surfaces do not establish biological relevance or affinity.

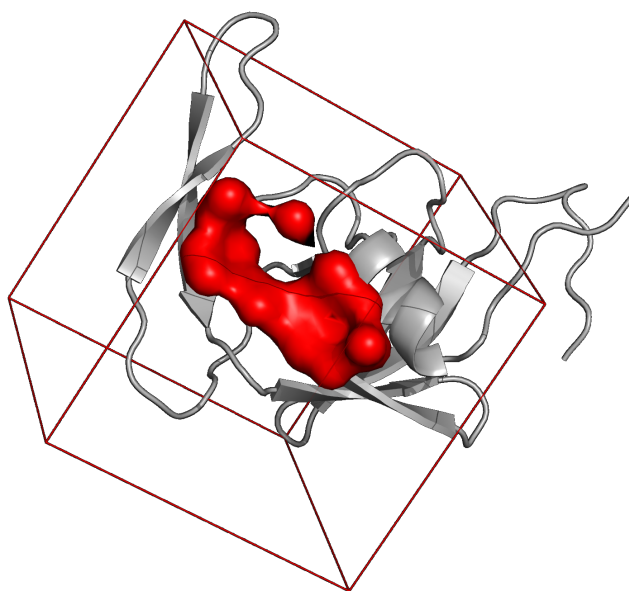


Figure 2. Selected docking region and box. All regions selected for docking are shown together on the receptor. Each pocket surface and its corresponding box use the same color assigned in the preceding A/B pocket-analysis figure and the related ligand-evidence figures. This records the regions chosen to proceed; it does not establish that the pockets are biologically correct.

3. Reproducibility, software, and references

fpocket generated the cavity candidates and descriptors. Receptor preparation: strict Meeko succeeded; PDBFixer was not needed. No docking or interaction analysis was performed. PyMOL, Matplotlib, and ReportLab generated this report.

Software versions used for this report

Result element	Software	Version used
Scientific workflow	Docking Universal	0.7.0
Ligand-free cavity detection	fpocket	4.2.2
Receptor preparation	Meeko	0.7.1
3D rendering	PyMOL	3.0.0
Plots	Matplotlib	3.9.4
PDF generation	ReportLab	5.0.0
Scientific workflow runtime	Python	3.9.23

Scientific and software references

1. Le Guilloux V, Schmidtke P, Tuffery P. Fpocket: an open source platform for ligand pocket detection. BMC Bioinformatics. 2009;10:168. <https://doi.org/10.1186/1471-2105-10-168>
2. Santos-Martins D, He Y, Eberhardt J, et al. Meeko: molecule parameterization and software interoperability for docking and beyond. J Chem Inf Model. 2025;65:13045-13050. <https://doi.org/10.1021/acs.jcim.5c02271>
3. The PyMOL Molecular Graphics System, Version 3.0.0, Schrodinger, LLC. <https://www.pymol.org/support.html>
4. Hunter JD. Matplotlib: A 2D Graphics Environment. Comput Sci Eng. 2007;9:90-95. <https://doi.org/10.1109/MCSE.2007.55>
5. ReportLab: PDF generation toolkit and user guide. <https://docs.reportlab.com/>
6. Python Software Foundation. Python Language Reference. <https://www.python.org/>